

Input & Generation

Prompt Tokens



Image Tokens



Current Policy

LLM Response



hidden
state

Visual Evidence Reasoning

Localization Module

Extract

visual signals

Compute

relevance score

Select

visual evidence

Reinforcement Module

Get

Evidence
Signals

Compute

Relative
Significance

Calculate

Token-level
Weight

Policy Update

Reward

[0, 2, 1, 3, ...]

Advantage

[-1, +0.5, +0.2, ...]

Advantage Shaping

[-1, +0.9, +0.3, ...]

optimize

New Policy